45V Tax Credit + Renewable PPA — ROI Analysis

Prepared: March 18, 2026 | **Source:** 45VH2-GREET (Rev. Dec 2025) + Argonne emission factors

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Executive Summary

Without clean power, Tobe does not qualify for 45V. With a 100% renewable PPA, each facility generates **$1.6M/year net benefit** ($16M over 10 years) after PPA costs.

Oklahoma Plains Region Grid Analysis

Source	Grid Share	Emission Factor	Weighted Contribution
Wind	36.6%	0 gCO2eq/kWh	0.0
Coal	31.8%	1,058 gCO2eq/kWh	336.4
Natural Gas	21.4%	487 gCO2eq/kWh	104.3
Nuclear	5.0%	2.5 gCO2eq/kWh	0.1
Hydro	4.3%	~0 gCO2eq/kWh	0.0
Biomass	0.3%	50.3 gCO2eq/kWh	0.2
Solar	0.3%	~0 gCO2eq/kWh	0.0
TOTAL	**100%**		**440.8 gCO2eq/kWh**

Source: GREET 2025, NEEDS Region "Plains"

Tobe Lifecycle Emissions by Electricity Source

Tobe electricity consumption: 42 kWh/kg H_2

Scenario	Grid %	Clean %	Lifecycle (kgCO2e/kg)	45V Credit	Annual Value*
Grid only	100%	0%	18.51	❌ None	$0

Scenario	Grid %	Clean %	Lifecycle (kgCO2e/kg)	45V Credit	Annual Value*
75% grid	75%	25%	13.88	❌ None	$0
50/50	50%	50%	9.26	❌ None	$0
25% grid	25%	75%	4.63	❌ None	$0
10% grid	10%	90%	1.85	$0.75/kg	$675K
5% grid	5%	95%	0.93	$1.00/kg	$900K
100% clean	**0%**	**100%**	**0.00**	**$3.00/kg** ⭐	**$2.7M**

*Per facility at 900,000 kg/year nameplate production

Renewable PPA Economics (100% Wind)

Cost Side

Item	Value	Source
Tobe electricity per facility	37.8 GWh/year	42 kWh/kg × 900,000 kg
Oklahoma wind PPA rate	$0.025–0.035/kWh	Market rates, 2025-2026
Annual PPA cost	$945K–$1.3M/year	37.8 GWh × rate
vs. grid electricity ($0.035/kWh)	$1.32M/year	Current Oklahoma industrial
Incremental PPA cost	**$0 – marginal**	Wind PPA may be at or below grid

Revenue Side (45V PTC)

Item	Value
Credit rate	$3.00/kg
Annual production	900,000 kg
Annual PTC revenue	**$2.7M**
Credit window	10 years
Lifetime PTC per facility	**$27M**

Net Benefit

Scenario	Annual PPA Cost	Annual PTC	Net Benefit	10-Year Net
PPA @ $0.025/kWh	$945K	$2.7M	**$1.76M/year**	**$17.6M**
PPA @ $0.030/kWh	$1.13M	$2.7M	**$1.57M/year**	**$15.7M**
PPA @ $0.035/kWh	$1.32M	$2.7M	**$1.38M/year**	**$13.8M**

Multi-Facility Impact (Financial Model Alignment)

Facilities	Annual PTC (total)	10-Year PTC (total)
1 (FY1-2)	$2.7M	$27M
3 (FY3)	$8.1M	$81M
4 (FY4+)	$10.8M	$108M
12 (FY7)	$32.4M	$324M

Financial model v8.5.3 assumes 4 facilities qualify (begin construction by Dec 2027)

Tobe's Efficiency Advantage

Electrolyzer	kWh/kg	MWh/year (900K kg)	PPA Cost @$0.03	Net after PTC
Tobe	**42**	**37,800**	**$1.13M**	**$1.57M**
Typical PEM	55	49,500	$1.49M	$1.21M
Typical ALK	60	54,000	$1.62M	$1.08M

Tobe's 42 kWh/kg efficiency means 24-30% less clean power needed than competitors to hit the same emissions threshold. Lower electricity bill + same PTC = wider margin.

Key Risks

1. **45V begin-construction deadline:** May be 2027 under OBBBA. Must break ground early.
2. **Hourly matching requirements:** Treasury may require temporal matching (not just annual RECs). Hourly-matched PPAs cost more.

3. **Additionality:** 45V may require NEW renewable capacity, not existing wind farms.
4. **Regulatory change:** Congress could modify/repeal. Model with and without.

Recommendation

Sign a 100% renewable wind PPA for every facility. The math is unambiguous: - Even at the highest PPA rate ($0.035), net benefit is $1.38M/year per facility - Oklahoma wind is abundant and cheap - Tobe's efficiency amplifies the advantage vs. competitors - The PTC alone could fund $13.8-17.6M per facility over 10 years

Next Steps: 1. Identify Oklahoma wind PPA providers (NextEra, AES, Invenergy) 2. Get indicative pricing for 10-year, 37.8 GWh/year contract 3. Confirm Treasury hourly matching requirements for 45V 4. Begin construction documentation before deadline

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Analysis by Ren ⚡ using 45VH2-GREET (Rev. December 2025) emission factors Argonne National Laboratory GREET model, Plains region, NEEDS electricity distribution